Patrick Amadeus Irawan

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Website LinkedIn Github

Research Summary

My research focuses on **multimodal foundation models and alignment** (training/test-time adaptation, image and video understanding), **agentic and action-conditioned LLM/VLM systems** (world models, reinforcement learning, sequential decision-making), and **large-scale evaluation** (multilingual and synthetic data benchmarking). My **pre-doctoral** experience spans both **academia and industry** in above topics, including contributions to award-winning and grant-funded research published at top-tier venues (**CVPR, *CL, EMNLP, COLING**). In practice, I primarily work on post-training (**SFT, preference optimization**) by utilizing action-conditioned supervision to build modality-grounded systems that go beyond language-only/biased priors.

Education

Ph.D. in Natural Language Processing — MBZUAI *2025 - 2029*
- Advisors: Alham Fikri Aji and Yova Kementchedjhieva
- *multimodal alignment; VLM/LLM; post-training*

Global Study Program — University of California, Davis *Jul 2023 - Dec 2023*
- *computational neuroscience, tech management*

B.Eng. in Computer Science — Institut Teknologi Bandung *2020 - 2024*
- *visual question answering; synthetic data generation; LLM reasoning*

Relevant Publications

LinguDistill: Recovering Linguistic Ability in Vision Language Models via Selective Cross-Modal Distillation [Preprint] — *Preprint* 2026
P. Irawan, E. Fuadi, S. Kumar, A. F. Aji, Y. Kementchedjhieva

Vision Language Models are Confused Tourists [Paper] — CVPR 2026
P. Irawan, I. Hanif, M. Kautsar, G. Winata, F. Koto, A. F. Aji

A Massive-Scale Multilingual Multi-Cultural Multimodal RAG [Paper] — CVPR 2026
D. Anugraha, P. Irawan, A. Singh, E. Lee, G. Winata

Counting to Four is Still a Chore For VLMs [TBA] — MMFM @ CVPR 2026
A. Duy, P. Irawan, T. Vo

Entropy2Vec: Crosslingual Language Modeling Entropy as End-to-End Learnable Language Representations [Paper] — MRL @ EMNLP 2025
P. Irawan, R. Diandaru*, B. Jagad*, R. Suchrady*, A. F. Aji, G. Winata, F. Koto, S. Cahyawijaya**

WorldCuisines: A Massive-Scale Benchmark for Multilingual and Multicultural VQA on Global Cuisines [Paper/Code] — NAACL 2025 (Best Theme Paper 🏆)
G. Winata, F. Hudi*, P. Irawan*, D. Anugraha*, R. Putri*, (20+ authors), et al.*

Seeing Culture: A Benchmark for Visual Reasoning and Grounding [Paper/Code] — EMNLP 2025
B. Satar, Z. Ma, P. Irawan, W. Mulyawan, J. Jiang, E. Lim, C.-W. Ngo

ProxyLM: Predicting Language Model Performance on Multilingual Tasks via Proxy Models [Paper] — NAACL 2025
D. Anugraha, G. Winata, C. Li, P. Irawan, E. Lee

Towards Efficient and Robust VQA-NLE Data Generation with Large Vision-Language Models [Paper] — COLING 2025
P. Irawan, G. Winata, S. Cahyawijaya, A. Purwarianti

SEACrowd: A Multilingual Multimodal Data Hub and Benchmark Suite for Southeast Asian Languages [Paper] EMNLP 2024
H. Lovenia, R. Mahendra*, S. Akbar*, L. Miranda*, (10+ authors), P. Irawan, et al.*

Work Experience

Research Engineer *Singapore*
Singapore Management University ***Feb 2025 – Sep 2025***

- Led a large-scale evaluation multimodal reasoning project, managing 5 collaborators (junior researchers, externals).
- Authored an EMNLP-accepted paper on segmentation methods for cultural and masked images.
- Continuing research on fine-grained novel video instance segmentation and moment localization in a large-scale, multicultural domain.

Software Engineer *Germany, Remote*
IT Bauschmiede ***Nov 2024 – Jul 2025***

- Part a core logistics team involving development on real-time scheduling, automated invoicing, and geolocation features, serving 5+ enterprise clients.
- Migrated system-wide database and refactored backend MVC into microservices, reducing overall processing time by ∼50% and enhancing reliability

Data Scientist Intern *Jakarta, ID*
Supertype ***Jan – Jul 2023***

- Designed & implemented topic extraction pipeline in conjunction with sentiment & semantic analysis, improving processing speed by 5x and reducing extracted topics' redundancy & inaccuracies by 33%
- Optimized Django REST + GCP backend via code refactoring, ensuring ∼95%+ uptime and increased throughput

Software Engineer Intern *Jakarta, ID*
Blibli ***Jun – Dec 2022***

- Expanded 4 commerce checkout services coverage via BDD, producing 150+ novel or improved QA cases in 6 months
- Optimized Kafka pipeline by implementing distributed programming, boosting test efficiency by 2x

Software Engineer Intern *Jakarta, ID*
Ruangguru ***Jan – Apr 2022***

- Developed a coding judge software for 1k+ users using Golang, whose smart scheduling & reminder features boosted task completion rate by ∼15% compared to legacy tool
- Eliminated code smell and enforced updated design pattern via Golang semantics & templates

Achievement & Awards

OpenAI-MBZUAI Micro Grant Award **Grantee**	*2025*
NAACL Best Theme Paper Award **Winner**	*2025*
BCG SEA Emeralds Case Competition **Winner**	*2024*
Indonesian International Student Mobility Awards @ UC Davis **Awardee**	*2023*
DOMO Higher Education Case Competition **Finalist**	*2023*
CSLeaders Full Bachelor Tuition Scholarship **Awardee**	*2022*
Gemastik XV Data Mining Division **Finalist**	*2022*

Technical Skills

Core Research: Python, PyTorch, Huggingface, OpenCV, CUDA, Langchain, Langsmith
Engineering & Infrastructure: Golang, Java, TypeScript, FastAPI, Django, REST, Flask, Gin, Echo, GCP
Supporting Tools / Frameworks: Looker Studio, Tableau, PowerBI, Data Analysis Library (Pandas, Numpy, Seaborn, etc.), Slurm, Bash